



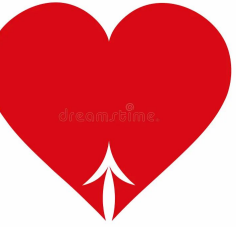
Poker Player Profiling: Heuristics vs. K-Means Clustering

Grouping Texas Hold'em players by how they bet.

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Introduction



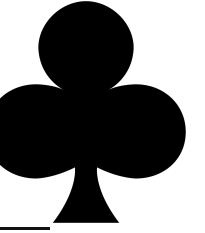
Goal

- Sort poker players into groups based on how they bet
- Poker players have different “styles” of play; useful for modeling opponents
- Past Approach: Use fixed cut offs on VPIP and PFR (two statistics) to categorize players
- Our Approach: Explore whether K-Means finds better groups via clustering

Dataset

- IRC Poker Database from the University of Alberta
 - Over 10 million Texas Hold'em hands played on IRC servers between 1995 and 2001

Related Works



IRC Poker Database (University of Alberta)

- The public hand-history dataset we collected data from
 - poker.cs.ualberta.ca/ircpokerdatabase.html

Categorizing Players Using VPIP/PFR Heuristics

- Long-standing practice in the poker community
- Players are sorted into TAG, LAG, Fish, and Rock using fixed ranges given by calculating different statistics
- This is our **ground truth**

K-Means clustering

- Standard unsupervised method for grouping similar items
- We apply it to players' VPIP and PFR values and compare it to the heuristic



Methodology: Data & Features



Steps

- Find every player log file across the human Hold'em channels:
 - **83,815 files**
- Parse each line into a record: player, hand id, position, and the action string for each round
- From the preflop action string, compute two flags per hand
 - **VPIP**: percentage of hands in which player voluntarily puts money into the pot preflop
 - How “tightly” a person plays
 - **PFR**: percentage of hands player makes preflop raise
- Add up flags per player to get VPIP and PFR percentages
- Keep only player with at least 30 hands
- Standardize VPIP and PFR before clustering

Filtering

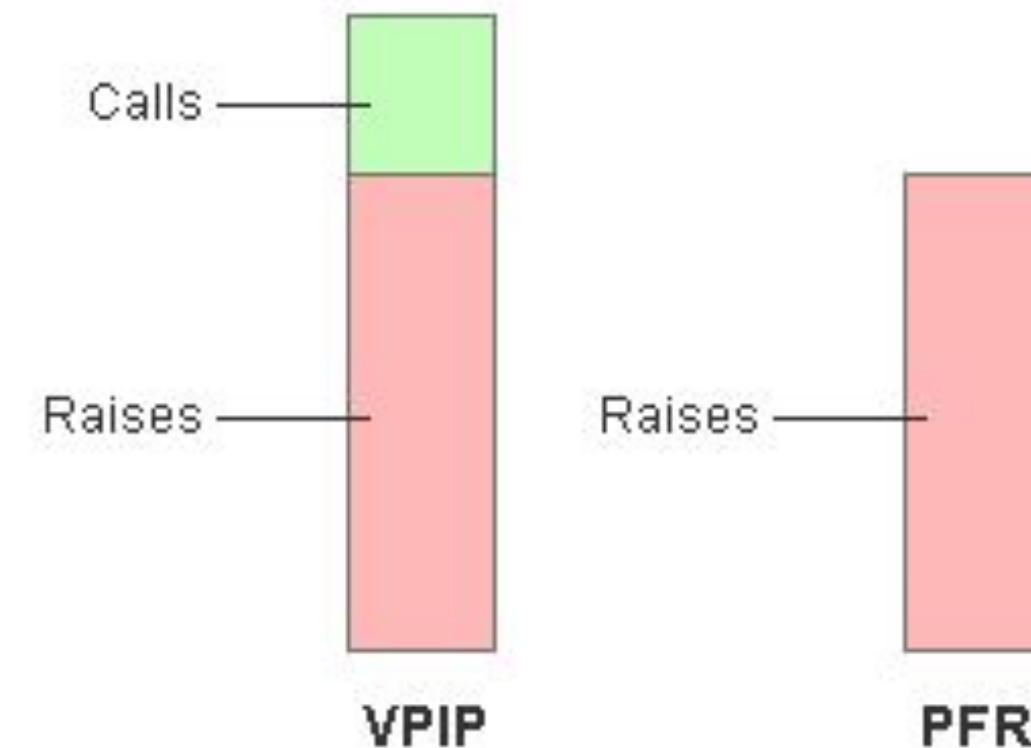
Players before filter: **24,786**

Removed (less than 30 hands): **8,347**

Players used for clustering: **16,439**

Channels included

- holdem, holdem1, holdemii



Methodology: Two Approaches



A) Heuristic baseline

- Fixed cut-offs: VPIP = 25%, PFR = 15%.

VPIP	< 25%	Tight
	>= 25%	Loose
PFR	< 15%	Passive
	>= 15%	Aggressive

- Each player goes into one of four categories:
 - **TAG:** tight, aggressive
 - **LAG:** loose, aggressive
 - **Fish:** loose, passive
 - **Rock:** tight, passive
- Same cutoff applies to everyone

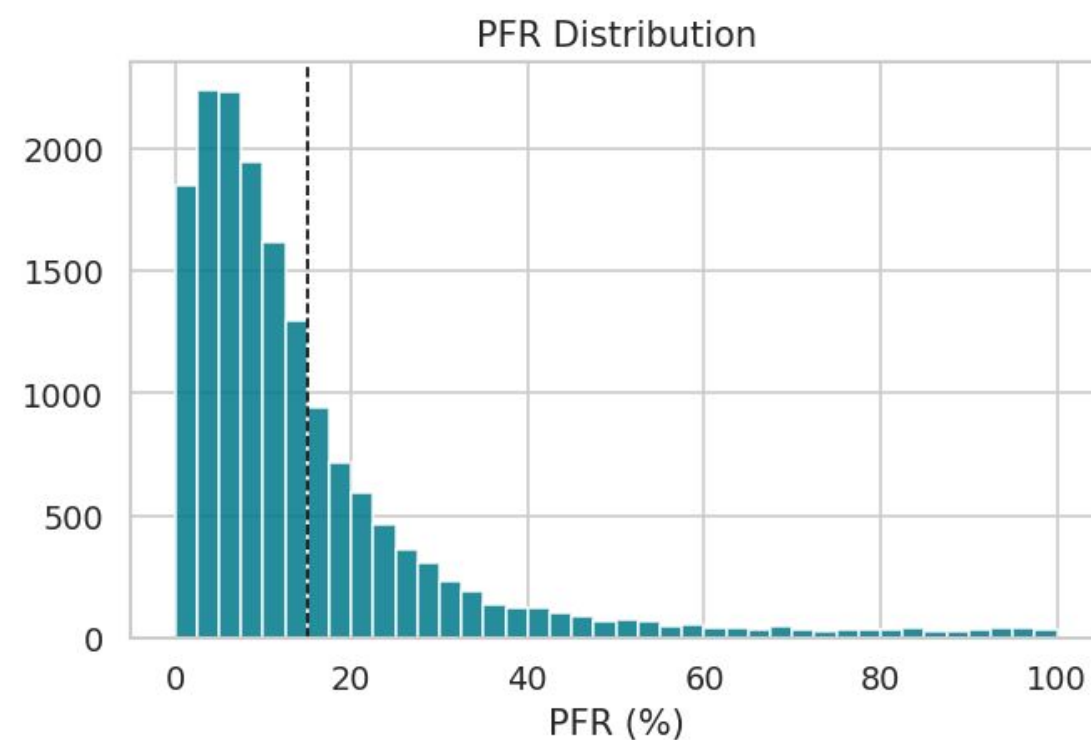
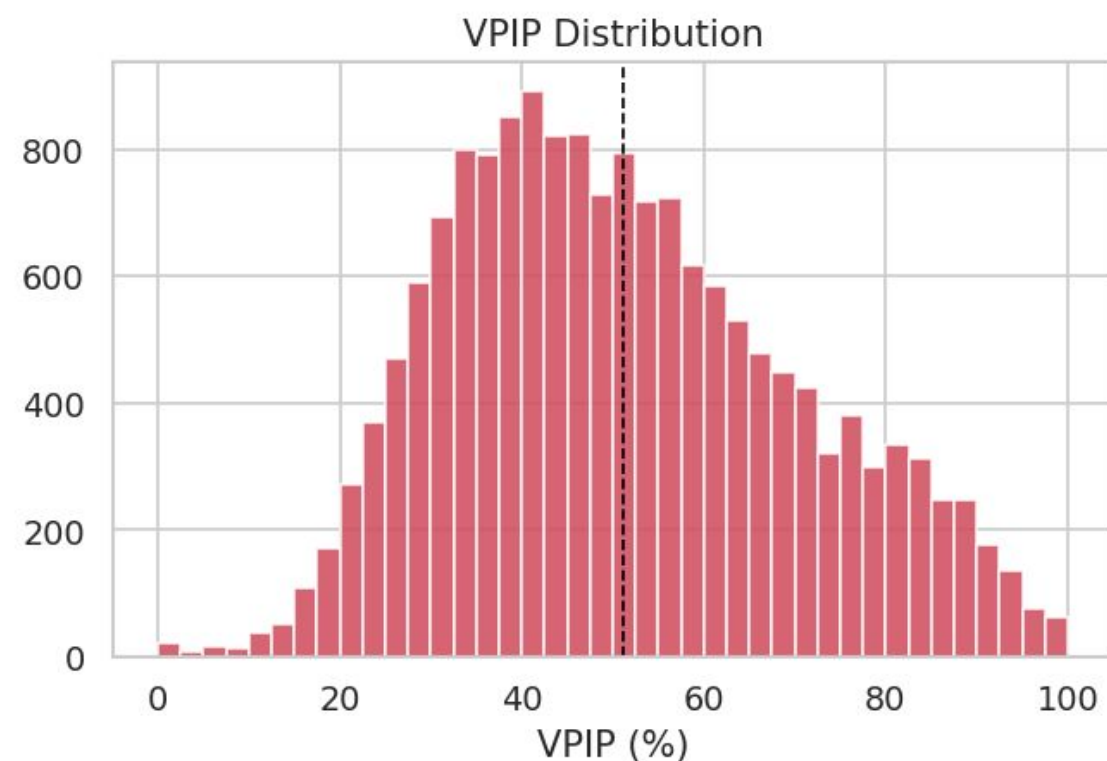
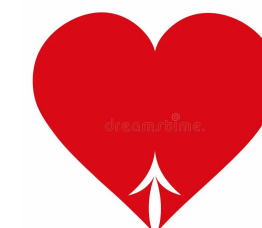
B) K-Means clustering

- Splits players into K groups by minimizing within-group distance

$$J = \sum_k \sum_{x \in C_k} \|x - \mu_k\|^2$$

- Final model uses **K = 4** to match the four poker archetypes
- Also tested K = 2 to 8 to check what the data prefers
 - elbow method, silhouette score
- Built with scikit-learn
 - **n_init = 20** (Run 20 consecutive times with random centroid initialization)
- Cluster quality measured by Silhouette Score (range -1 to +1; higher is better)

Results: Feature Distributions



Quick stats (n = 16,439)

- VPIP: mean **51.08%**, std **19.14**
- PFR: mean **15.01%**, std **16.64**
- Hands per player: median **225**, max **280,960**
- Correlation VPIP vs PFR: **0.485**

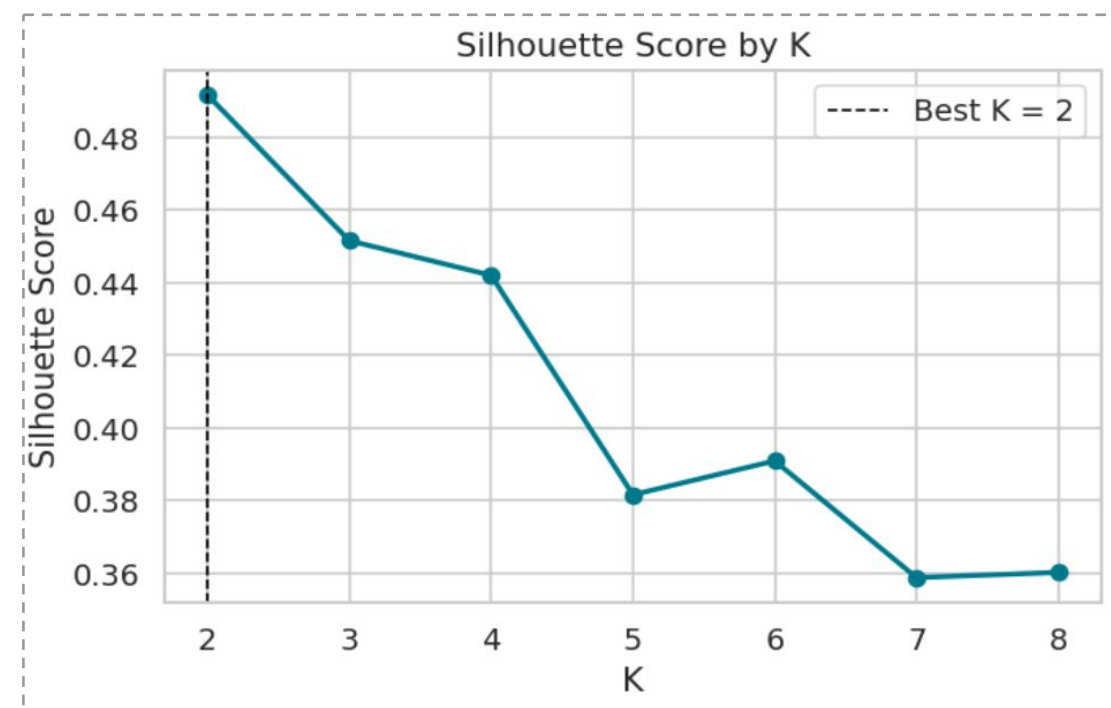
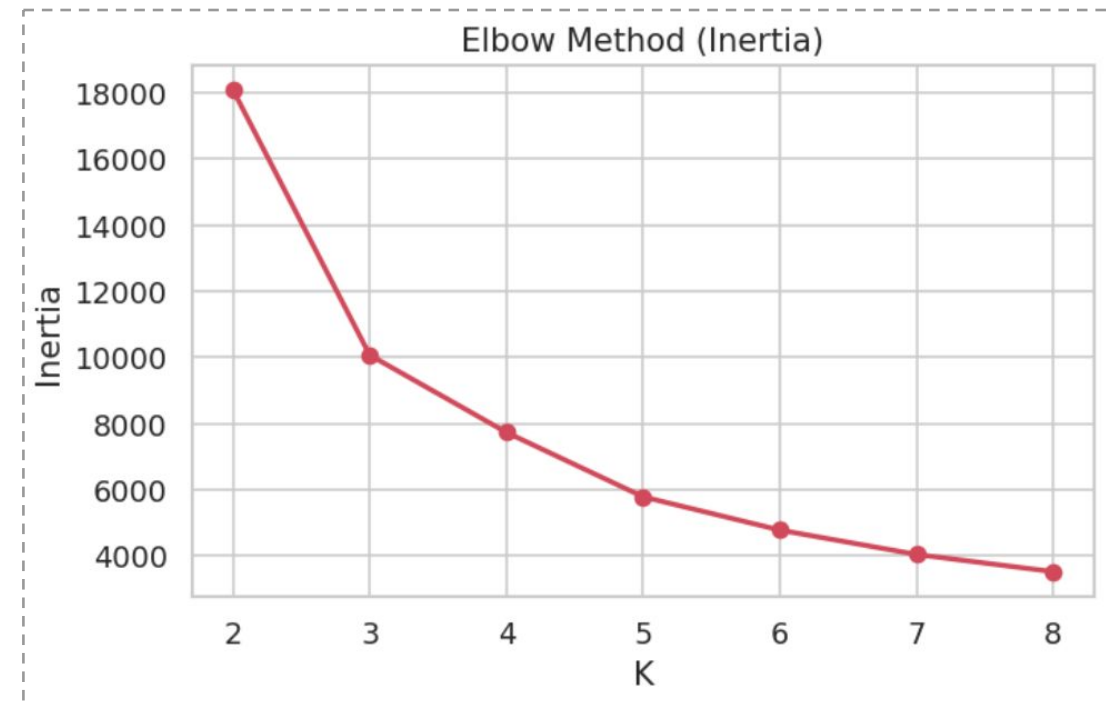
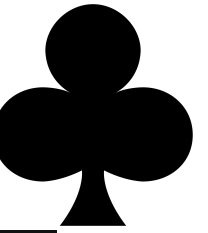
Sanity checks

- PFR > VPIP violations: **0** (should be 0)
- Duplicate player rows: **0**

Heuristic label counts

- Fish: **10,140** · LAG: **5,225**
- Rock: **1,042** · TAG: **32**

Results: Choosing K



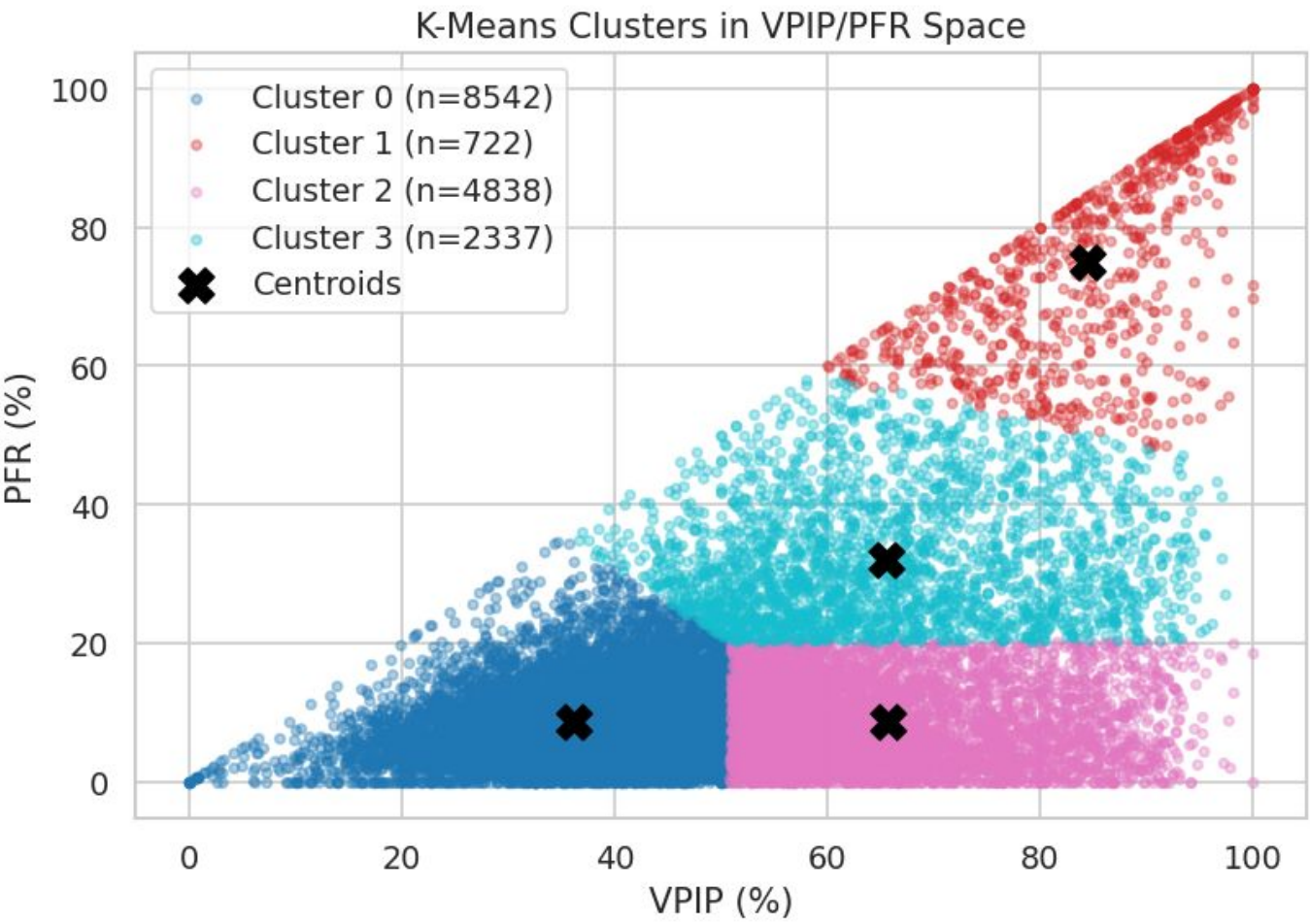
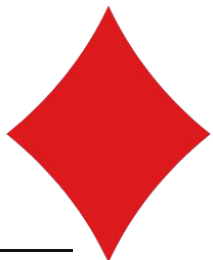
Scores for K = 2 to 8

K	Inertia	Silhouette
2	18,086.23	0.4917
3	10,067.76	0.4516
4	7,730.94	0.4420
5	5,781.48	0.3816
6	4,775.33	0.3909
7	4,038.98	0.3588
8	3,518.03	0.3602

What we picked

- Best by silhouette: **K = 2** (0.4917)
- Final reported model: **K = 4** (0.4420): to match the four poker archetypes
- Scores drop after K = 5, more groups don't help

Results: Final Clusters (K = 4)



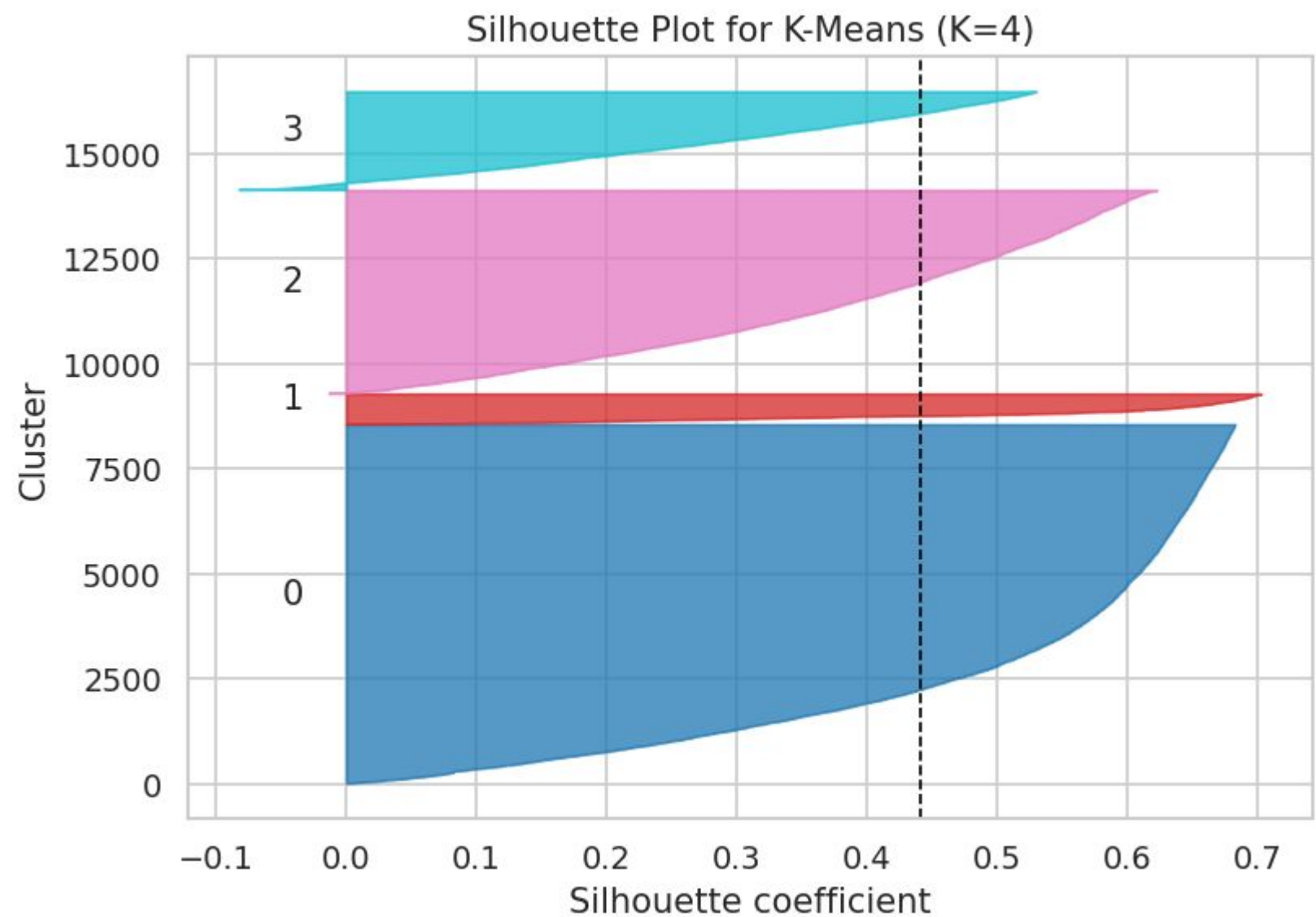
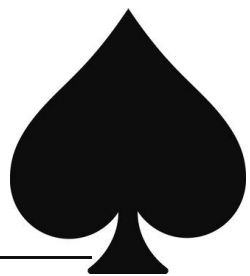
Cluster centers and sizes

Cluster	Size	VPIP	PFR
0	8,542	36.11	8.88
1	722	84.39	74.86
2	4,838	65.65	8.74
3	2,337	65.59	32.00

Cluster vs heuristic label

Cluster	Fish	LAG	Rock	TAG
0	6,150	1,318	1,042	32
1	0	722	0	0
2	3,990	848	0	0
3	0	2,337	0	0

Analysis: K-Means vs. Heuristic



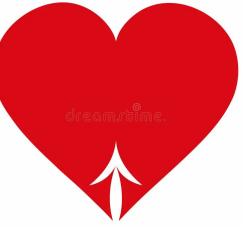
Silhouette scores

Method	Groups	Score
Heuristic baseline	4	0.1347
K-Means (K = 4)	4	0.4420
K-Means (best K, K = 2)	2	0.4917

What this tells us

- K-Means at K = 4 produces tighter, better separated groups than the heuristic
- The heuristic's "LAG" label gets split across three K-Means clusters and the data has more structure than fixed cutoffs can show
- The same direction holds on the smaller sampled subset (n = 344)
 - K-Means K = 4 silhouette 0.4595 vs heuristic 0.0990

Conclusion & Future Work



Conclusion

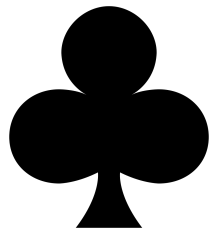
- K-Means on standardized VPIP and PFR gives tighter groups than fixed-threshold rules
 - At $K = 4$, silhouette is **0.4420** (K-Means) vs **0.1347** (heuristic)
- The data's strongest natural split is binary ($K = 2$, silhouette **0.4917**)
 - K-Means beats the heuristic on the project's main metric

Limitations

- Only preflop actions are used; flop, turn, and river are ignored
- Each player is one static average over their full history
 - Data is from 1995–2001; modern poker differs
 - K-Means assumes round, equal size clusters

Future work

- Add postflop features (3-bet %, c-bet %, etc.)
- Try a more recent dataset
 - Before and after economic turmoil and rise of GTO play
- Try other clustering methods that don't assume round clusters



Thank You!

Card matrix

AA	AKs	AQs	AJs	ATs	A9s	A8s	A7s	A6s	A5s	A4s	A3s	A2s
AKo	KK	KQs	KJs	KTs	K9s	K8s	K7s	K6s	K5s	K4s	K3s	K2s
AQo	KQo	QQ	QJs	QTs	Q9s	Q8s	Q7s	Q6s	Q5s	Q4s	Q3s	Q2s
AJo	KJo	QJo	JJ	JTs	J9s	J8s	J7s	J6s	J5s	J4s	J3s	J2s
ATo	KTo	QTo	JTo	TT	T9s	T8s	T7s	T6s	T5s	T4s	T3s	T2s
A9o	K9o	Q9o	J9o	T9o	99	98s	97s	96s	95s	94s	93s	92s
A8o	K8o	Q8o	J8o	T8o	98o	88	87s	86s	85s	84s	83s	82s
A7o	K7o	Q7o	J7o	T7o	97o	87o	77	76s	75s	74s	73s	72s
A6o	K6o	Q6o	J6o	T6o	96o	86o	76o	66	65s	64s	63s	62s
A5o	K5o	Q5o	J5o	T5o	95o	85o	75o	65o	55	54s	53s	52s
A4o	K4o	Q4o	J4o	T4o	94o	84o	74o	64o	54o	44	43s	42s
A3o	K3o	Q3o	J3o	T3o	93o	83o	73o	63o	53o	43o	33	32s
A2o	K2o	Q2o	J2o	T2o	92o	82o	72o	62o	52o	42o	32o	22

PFR: 19.9 VPIP: 50.0

Suit selection

Clear matrix